math

Justin Lee

October 2026

1 Softmax Backward

Start from the chain rule:

$$\frac{\partial L}{\partial x_j} = \sum_i \frac{\partial L}{\partial s_i} \frac{\partial s_i}{\partial x_j}$$

Define:

$$g_i = \frac{\partial L}{\partial s_i}$$

And plug in:

$$\frac{\partial s_i}{\partial x_j} = s_i(\delta_{ij} - s_j)$$

So:

$$\frac{\partial L}{\partial x_j} = \sum_i g_i s_i (\delta_{ij} - s_j)$$

Expand:

$$= \sum_i g_i s_i \delta_{ij} - \sum_i g_i s_i s_j$$

Now, δ_{ij} is 1 only when $i = j$, so:

$$\sum_i g_i s_i \delta_{ij} = g_j s_j$$

And s_j does not depend on the summation index i, so:

$$\sum_i g_i s_i s_j = s_j \sum_i g_i s_i$$

Therefore:

$$\frac{\partial L}{\partial x_j} = g_j s_j - s_j \sum_i g_i s_i$$

Factor out s_j:

$$\boxed{\frac{\partial L}{\partial x_j} = s_j \left(g_j - \sum_i g_i s_i \right)}$$

2 LayerNorm Backward

2.1 Forward Equations

For one token with hidden dimension H:

$$\mu = \frac{1}{H}\sum_j x_j$$

$$\sigma^2 = \frac{1}{H}\sum_j (x_j - \mu)^2$$

Define:

$$r = \frac{1}{\sqrt{\sigma^2 + \epsilon}}$$

The normalized input is:

$$\hat{x}_i = (x_i - \mu)r$$

The LayerNorm output is:

$$y_i = \gamma_i \hat{x}_i + \beta_i$$

Define the upstream gradient:

$$dy_i = \frac{\partial L}{\partial y_i}$$

2.2 Gradients of Gamma and Beta

From:

$$y_i = \gamma_i \hat{x}_i + \beta_i$$

We have:

$$\frac{\partial y_i}{\partial \beta_i} = 1, \qquad \frac{\partial y_i}{\partial \gamma_i} = \hat{x}_i$$

Therefore, for a single token:

$$d\beta_i = dy_i$$

$$d\gamma_i = dy_i \hat{x}_i$$

Since gamma and beta are shared across all tokens, we sum the gradients over all rows:

$$\boxed{d\beta_i = \sum_{\text{rows}} dy_i}$$

$$\boxed{d\gamma_i = \sum_{\text{rows}} dy_i \hat{x}_i}$$

2.3 Gradient of Input

Define:

$$g_i = d\hat{x}_i = \frac{\partial L}{\partial \hat{x}_i} = dy_i \gamma_i$$

We want:

$$dx_i = \frac{\partial L}{\partial x_i}$$

There are three gradient paths from x_i to the loss.

2.3.1 Path 1: Direct Path

Since:

$$\hat{x}_i = (x_i - \mu) r$$

Holding μ and r fixed:

$$\frac{\partial \hat{x}_i}{\partial x_i} = r$$

Therefore:

$$\boxed{dx_i^{(1)} = g_i r}$$

2.3.2 Path 2: Through the Mean

Recall:

$$\mu = \frac{1}{H} \sum_j x_j$$

By the chain rule:

$$\frac{\partial L}{\partial \mu} = \sum_j \frac{\partial L}{\partial \hat{x}_j} \frac{\partial \hat{x}_j}{\partial \mu}$$

Since:

$$\hat{x}_j = (x_j - \mu) r$$

Holding r fixed:

$$\frac{\partial \hat{x}_j}{\partial \mu} = -r$$

Thus:

$$\frac{\partial L}{\partial \mu} = \sum_j g_j (-r) = -r \sum_j g_j$$

Also:

$$\frac{\partial \mu}{\partial x_i} = \frac{1}{H}$$

Therefore:

$$dx_i^{(2)} = \frac{\partial L}{\partial \mu} \frac{\partial \mu}{\partial x_i}$$

$$\boxed{dx_i^{(2)} = -\frac{r}{H} \sum_j g_j}$$

2.3.3 Path 3: Through the Variance

Recall:

$$\sigma^2 = \frac{1}{H}\sum_j (x_j - \mu)^2$$

And:

$$r = (\sigma^2 + \epsilon)^{-1/2}$$

Differentiate r with respect to σ^2:

$$\frac{\partial r}{\partial \sigma^2} = -\frac{1}{2}(\sigma^2 + \epsilon)^{-3/2}$$

Since:

$$r^3 = (\sigma^2 + \epsilon)^{-3/2}$$

We get:

$$\frac{\partial r}{\partial \sigma^2} = -\frac{1}{2}r^3$$

Now apply the chain rule:

$$\frac{\partial L}{\partial \sigma^2} = \sum_j \frac{\partial L}{\partial \hat{x}_j}\frac{\partial \hat{x}_j}{\partial \sigma^2}$$

Since:

$$\hat{x}_j = (x_j - \mu)r$$

Holding μ fixed:

$$\frac{\partial \hat{x}_j}{\partial \sigma^2} = (x_j - \mu)\frac{\partial r}{\partial \sigma^2}$$

$$= -\frac{1}{2}r^3(x_j - \mu)$$

Therefore:

$$\boxed{\frac{\partial L}{\partial \sigma^2} = -\frac{1}{2}r^3\sum_j g_j(x_j - \mu)}$$

Next, differentiate the variance:

$$\frac{\partial \sigma^2}{\partial x_i} = \frac{2(x_i - \mu)}{H}$$

The terms involving the derivative of μ cancel because $\sum_j (x_j - \mu) = 0$.
By the chain rule:

$$dx_i^{(3)} = \frac{\partial L}{\partial \sigma^2}\frac{\partial \sigma^2}{\partial x_i}$$

Substitute:

$$dx_i^{(3)} = \left(-\frac{1}{2}r^3\sum_j g_j(x_j - \mu)\right)\frac{2(x_i - \mu)}{H}$$

Cancel the factor of 2:

$$dx_i^{(3)} = -\frac{r^3}{H}(x_i - \mu)\sum_j g_j(x_j - \mu)$$

Since:

$$\hat{x}_i = (x_i - \mu)r$$

We have:

$$x_i - \mu = \frac{\hat{x}_i}{r}$$

Substitute:

$$dx_i^{(3)} = -\frac{r^3}{H}\frac{\hat{x}_i}{r}\sum_j g_j\frac{\hat{x}_j}{r}$$

Simplify:

$$\boxed{dx_i^{(3)} = -\frac{r}{H}\hat{x}_i\sum_j g_j\hat{x}_j}$$

2.4 Final Input Gradient

Add the three gradient paths:

$$dx_i = dx_i^{(1)} + dx_i^{(2)} + dx_i^{(3)}$$

$$dx_i = g_i r - \frac{r}{H}\sum_j g_j - \frac{r}{H}\hat{x}_i\sum_j g_j\hat{x}_j$$

Factor out r:

$$\boxed{dx_i = r\left[g_i - \frac{1}{H}\sum_j g_j - \frac{\hat{x}_i}{H}\sum_j g_j\hat{x}_j\right]}$$

Recall:

$$g_i = d\hat{x}_i = dy_i\gamma_i$$

And:

$$r = \frac{1}{\sqrt{\sigma^2 + \epsilon}}$$

Therefore, the final LayerNorm backward formula is:

$$\boxed{dx_i = \frac{1}{\sqrt{\sigma^2 + \epsilon}}\left[d\hat{x}_i - \frac{\sum_j d\hat{x}_j + \hat{x}_i\sum_j d\hat{x}_j\hat{x}_j}{H}\right]}$$